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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2015 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 2

Friday 20th February 2015

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 55 Marks

- All questions may be attempted.

Section I – 7 Marks

- Questions 1–7 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eight.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 73 boys

Examiner

MLS

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which of the following represents the complex number $\sqrt{3} - i$ expressed in modulus–argument form?

1

- (A) $2 \operatorname{cis} \frac{\pi}{6}$
- (B) $2 \operatorname{cis} \left(-\frac{\pi}{6}\right)$
- (C) $2 \operatorname{cis} \frac{2\pi}{3}$
- (D) $2 \operatorname{cis} \left(-\frac{2\pi}{3}\right)$

QUESTION TWO

What are the solutions to the quadratic equation $z^2 - 4iz + 12 = 0$?

1

- (A) $-6i$ or $2i$
- (B) $6i$ or $-2i$
- (C) $2 + 2\sqrt{2}i$ or $2 - 2\sqrt{2}i$
- (D) $2i + 2\sqrt{2}$ or $2i - 2\sqrt{2}$

QUESTION THREE

Which of the following represents the complex number $-iz$ where $|z| = 2\sqrt{2}$ and $\arg(z) = \frac{\pi}{4}$?

1

- (A) $2(1 - i)$
- (B) $2(1 + i)$
- (C) $2(-1 + i)$
- (D) $2(-1 - i)$

Exam continues next page ...

QUESTION FOUR

The expression $\frac{9x}{(x+2)(x-1)^2}$ can be expressed in partial fractions as:

1

(A) $\frac{2}{x-1} - \frac{2}{x+2} + \frac{3}{(x-1)^2}$

(B) $\frac{2}{x+2} - \frac{3}{(x-1)^2}$

(C) $\frac{2}{x+2} + \frac{3}{(x-1)^2}$

(D) $\frac{2}{x-1} + \frac{2}{x+2} + \frac{3}{(x-1)^2}$

QUESTION FIVE

Suppose z and w are the roots of the quadratic equation $3x^2 + (2-i)x + (4+i) = 0$. Without solving the quadratic find the value of $\bar{z} + \bar{w}$.

1

(A) $-\frac{2+i}{3}$

(B) $\frac{2+i}{3}$

(C) $-\frac{4+i}{3}$

(D) $\frac{4+i}{3}$

QUESTION SIX

Which of the following is a primitive of $\frac{x-1}{x^2+4}$?

1

(A) $2 \log_e(x^2+4) - 2 \tan^{-1} x + C$

(B) $\frac{1}{2} \log_e(2x) - \tan^{-1} x + C$

(C) $2 \log_e(x^2+4) - \tan^{-1} \frac{x}{2} + C$

(D) $\frac{1}{2} \log_e(x^2+4) - \frac{1}{2} \tan^{-1} \frac{x}{2} + C$

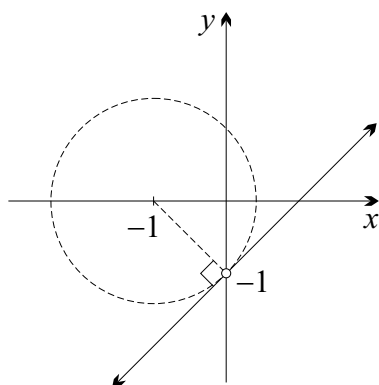
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QUESTION SEVEN

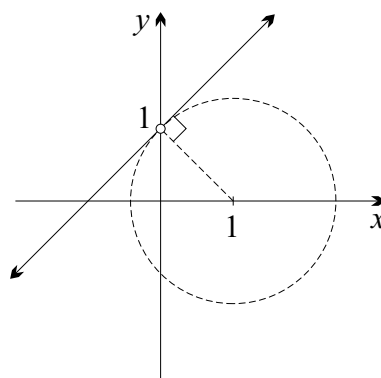
If $w = \frac{z+1}{z+i}$ and w is imaginary, what is the locus of z ?

1

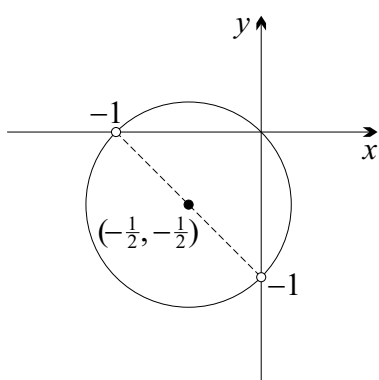
(A)



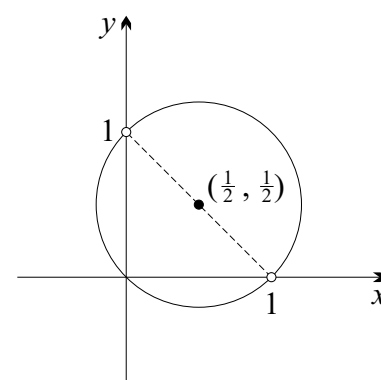
(B)



(C)



(D)



————— End of Section I —————

Exam continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION EIGHT (12 marks) Use a separate writing booklet.	Marks
(a) Let $u = 8 - 2i$ and $w = 4 + 3i$.	
(i) Find $\text{Im}(uw)$.	1
(ii) Find $u + \overline{w}$.	1
(iii) Express $\frac{u}{w}$ in the form $a + ib$, where a and b are real numbers.	2
(b) (i) Describe the locus of those points z such that $ z - i = z + i $.	1
(ii) Describe and sketch the locus of those points z such that $ z - i = \sqrt{2} z + i $. Do not find any intercepts with the axes.	3
(c) (i) Find all pairs of integers x and y such that $(x + iy)^2 = -3 - 4i$.	2
(ii) Using part (i), or otherwise, solve the quadratic equation $z^2 - 3z + (3 + i) = 0$.	2

QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

(a) Evaluate $\int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\cos^2 x} dx$. **2**

(b) Evaluate $\int_3^8 \frac{x}{(x+1) - \sqrt{x+1}} dx$ by using the substitution $x+1 = u^2$. **3**

(c) (i) Find A and B such that $\frac{6x+23}{2x^2+11x-6} = \frac{A}{2x-1} + \frac{B}{x+6}$. **2**

(ii) Find $\int \frac{6x+23}{2x^2+11x-6} dx$. **1**

(d) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{5+3\sin x+4\cos x}$. **4**

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

- (a) A curve has parametric equations

3

$$x = \sin \theta$$

$$y = \tan \theta$$

where $0 < \theta \leq \pi$ and $\theta \neq \frac{\pi}{2}$.

Show that $\frac{dy}{dx} = \sec^3 \theta$ and $\frac{d^2y}{dx^2} = 3 \tan \theta \sec^4 \theta$.

- (b) Using the substitution $x = 6 \cos u$, determine $\int \frac{1}{x^2 \sqrt{36 - x^2}} dx$.

3

- (c) Suppose the locus of a point P in the Argand diagram has equation $|z - 2i| = 1$.

(i) Sketch the locus of P as z varies.

1

(ii) Find the maximum and minimum values of $\arg z$, if $0 < \arg z < \pi$.

3

(iii) Find the value of z when $\arg z$ takes the minimum value found in part (ii).
Give z in the form $a + ib$.

2

QUESTION ELEVEN (12 marks) Use a separate writing booklet.**Marks**

(a) Use integration by parts to determine $\int x^2 \cos x \, dx$. **2**

(b) You are given that

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B).$$

Let $S = 1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta$.

(i) Show that $S \times \sin \frac{\theta}{2} = \sin \frac{7\theta}{2}$. **1**

(ii) Hence show that $1 + 2 \cos \frac{2\pi}{7} + 2 \cos \frac{4\pi}{7} + 2 \cos \frac{6\pi}{7} = 0$. **1**

(iii) By writing S as a sum of powers of $\cos \theta$, show that $\cos \frac{2\pi}{7}$ is a solution of the polynomial equation $8x^3 + 4x^2 - 4x - 1 = 0$. **2**

(c) Suppose z is a complex number such that $|z - 1| = 1$ and $\text{Im}(z) > 0$. Let the points O, P and Q represent $0 + 0i$, z and $z - 1$ respectively.

(i) Draw a diagram showing O, P and Q . **1**

(ii) Let $\arg(z) = \theta$. Write expressions for $\arg(z - 1)$ and $\arg(z - 2)$ in terms of θ . **1**

(iii) Show that $\arg\left(\frac{z^2 - 2z}{z - 1}\right) = \frac{\pi}{2}$. **1**

(iv) What can we deduce about the locus of the point representing $\frac{z^2 - 2z}{z - 1}$ from the result in part (iii)? **1**

(v) Find the maximum value of $\left|\frac{z^2 - 2z}{z - 1}\right|$. **2**

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

MC

Q1. $\begin{array}{c} | \\ \hline | \\ \times \end{array}$ B.

Q2. $z^2 - 4iz + 12 = 0$
 $(z - 6i)(z + 2i) = 0$
 $z = 6i \text{ or } -2i$ B

Q3. $z = 2\sqrt{2} \cos \frac{\pi}{4}$
 $= 2\sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right)$
 $= 2 + 2i$
 $-iz = -i(2 + 2i)$
 $= 2 - 2i$
 $= 2(1 - i)$ A.

Q4. $\frac{9x}{(x+2)(x-1)^2} = \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$
 $9x = A(x-1)^2 + B(x+2)(x-1) + C(x+2)$
 $x=1$ $9 = 3C$, $C=3$
 $x=-2$, $-18 = 9A$, $A=-2$ A.

Q5. $3x^2 + (2-i)x + (4+i) = 0$
 $z + w = -\frac{(2-i)}{3}$

$\bar{z} + \bar{w} = \overline{z+w}$
 $= -\frac{(2+i)}{3}$ A.

6.

$$\int \frac{x-1}{x^2+4} dx = \frac{1}{2} \int \frac{2x}{x^2+4} dx - \int \frac{1}{x^2+4} dx \quad D$$

$$= \frac{1}{2} \ln|x^2+4| - \frac{1}{2} \tan^{-1} \frac{x}{2} + C$$

7. let $z = x+iy$

$$w = \frac{x+1+iy}{x+i(y+1)} \times \frac{x-i(y+1)}{x-i(y+1)}$$

$$= \frac{x(x+1) + iyx - i(y+1)(x+1) + y(y+1)}{x^2 + (y+1)^2}$$

imag if $\operatorname{re}(w) = 0$

ie $x(x+1) + y(y+1) = 0$

$$x^2+x + y^2+y = 0$$

$$x^2+x+\frac{1}{4} + y^2+y+\frac{1}{4} = \frac{1}{2}$$

$$\left(x+\frac{1}{2}\right)^2 + \left(y+\frac{1}{2}\right)^2 = \frac{1}{2}$$

circle centre $\left(-\frac{1}{2}, -\frac{1}{2}\right)$, radius $\frac{1}{\sqrt{2}}$.

Q8.

a)

$$(i) \quad u\omega = (8-2i)(4+3i) \\ = 32 - 8i + 24i + 6$$

$$\operatorname{Im}(u\omega) = 16 \quad \checkmark$$

$$(ii) \quad u + \bar{\omega} = 8-2i + 4-3i \\ = 12-5i \quad \checkmark$$

$$(iii) \quad \frac{u}{\omega} = \frac{8-2i}{4+3i} \times \frac{4-3i}{4-3i} \quad \checkmark \\ = \frac{32 - 8i - 24i - 6}{16+9} \\ = \frac{26 - 32i}{25} \quad \checkmark \\ = \frac{1}{25}(26 - 32i)$$

$$(b) (i) \quad |z-i| = |z+i|$$

We want the locus of all points
equidistant from i and $-i$

i.e. the x axis or real axis or $y=0$.

$\checkmark$

$$ii) |z-i| = \sqrt{2} |z+i|$$

$$\text{let } z = x+iy$$

$$|x+i(y-1)| = \sqrt{2} |x+i(y+1)|$$

$$x^2 + (y-1)^2 = 2(x^2 + (y+1)^2)$$

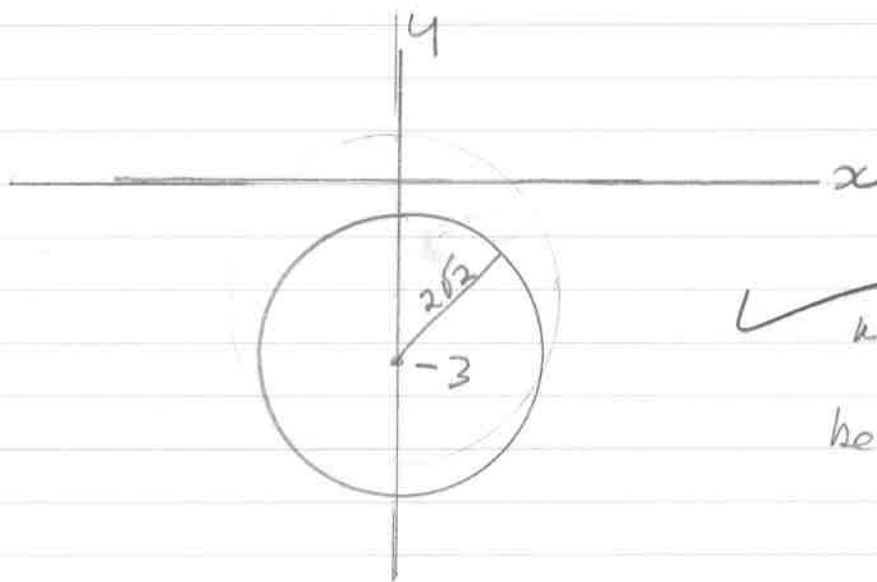
$$x^2 + y^2 - 2y + 1 = 2x^2 + 2y^2 + 4y + 2$$

$$x^2 + y^2 + 6y + 1 = 0$$

$$x^2 + (y^2 + 6y + 9) = -1 + 9$$

$$x^2 + (y+3)^2 = 8$$

circle centre $(0, -3)$, radius $2\sqrt{2}$.



✓ keep centre & radius to be clear.

c. (i) $(x+iy)^2 = -3-4i$

$$x^2 - y^2 + 2xyi = -3 - 4i$$

$$x^2 - y^2 = -3$$

$$xy = -2$$

$$y = -\frac{2}{x}$$

$$x^2 - \frac{4}{x^2} = -3$$

$$x^4 + 3x^2 - 4 = 0$$

$$(x^2 + 4)(x^2 - 1) = 0$$

$$x = 1 \text{ or } -1$$

$$y = -2 \text{ or } 2$$

$$(1-2i) \text{ or } (-1+2i)$$

(ii) $z^2 - 3z + (3+i) = 0$

$$z = \frac{3 \pm \sqrt{9 - 4(3+i)}}{2}$$

$$= \frac{3 \pm \sqrt{-3-4i}}{2}$$

$$= \frac{3 \pm (1-2i)}{2}$$

$$= \frac{4-2i}{2} \text{ or } \frac{2+2i}{2}$$

$$= (2-i) \text{ or } (1+i)$$

Q9.

$$(a) \int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\cos^2 x} dx = \int_0^{\frac{\pi}{4}} \sec^2 x e^{\tan x} dx \quad \checkmark$$

$$= \left[e^{\tan x} \right]_0^{\frac{\pi}{4}}$$

$$= e - 1 \quad \checkmark$$

$$(b) \int_3^8 \frac{x}{(x+1) - \sqrt{x+1}} dx$$

$$x+1 = u^2$$

$$dx = 2u du$$

x	8	3
u	3	2

$$= \int_2^3 \frac{u^2 - 1}{u^2 - u} 2u du \quad \checkmark$$

$$= 2 \int_2^3 \frac{(u+1)(u-1)}{u-1} du$$

$$= 2 \int_2^3 (u+1) du \quad \checkmark$$

$$= \left[u^2 + 2u \right]_2^3$$

$$= (9+6) - (4+4)$$

$$= 7 \quad \checkmark$$

c.

$$(i) \frac{6x+23}{2x^2+11x-6} = \frac{A}{2x-1} + \frac{B}{x+6}$$

$$x = -6, \quad \begin{array}{l} 6x+23 = A(x+6) + B(2x-1) \\ -13 = -13B \Rightarrow B = 1 \end{array} \quad \checkmark$$

$$x = \frac{1}{2}, \quad 26 = \frac{13}{2}A \Rightarrow A = 4 \quad \checkmark$$

$$(ii) \int \frac{6x+23}{2x^2+11x-6} dx = \int \frac{4}{2x-1} + \frac{1}{x+6} dx$$

$$= 2\ln(2x-1) + \ln(x+6) + C \quad \checkmark$$

(d)

$$\int_0^{\frac{\pi}{2}} \frac{dx}{5+3\sin x+4\cos x}$$

$$= \int_0^1 \frac{2dt}{5 + \frac{6t}{1+t^2} + \frac{4-4t^2}{1+t^2}} \times \frac{1}{(1+t^2)} \quad \checkmark$$

$$= \int_0^1 \frac{2dt}{5+5t^2+6t+4-4t^2}$$

$$= \int_0^1 \frac{2dt}{t^2+6t+9} \quad \checkmark$$

$$= 2 \int_0^1 (t+3)^{-2} dt$$

let $t = \tan \frac{x}{2}$

$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$

$= \frac{1}{2} (\tan^2 \frac{x}{2} + 1)$

$\frac{dt}{dx} = \frac{1}{2} (t^2 + 1)$

$dx = \frac{2dt}{1+t^2}$

x	$\frac{\pi}{2}$	0
t	1	0

$$= -2 \left[(t+3)^{-1} \right]_0^1 \checkmark$$

$$= -2 \left[\frac{1}{4} - \frac{1}{3} \right]$$

$$= -2 \times -\frac{1}{12}$$

$$= \frac{1}{6} \checkmark$$

Q10.

a). $x = \sin \theta$
 $\frac{dx}{d\theta} = \cos \theta$

$$y = \tan \theta$$
$$\frac{dy}{d\theta} = \sec^2 \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \frac{d\theta}{dx}$$
$$= \frac{\sec^2 \theta}{\cos \theta}$$

$$= \sec^3 \theta$$

$$\frac{d^2 y}{dx^2} = \frac{d(\sec^3 \theta)}{dx}$$

$$= \frac{d(\sec^3 \theta)}{d\theta} \times \frac{d\theta}{dx}$$

$$= 3 \sec^2 \theta \cdot \sec \theta \tan \theta \frac{d\theta}{dx}$$

$$= 3 \sec^3 \theta \tan \theta \frac{1}{\cos \theta}$$

$$= 3 \tan \theta \sec^4 \theta$$

} use of chain rule ✓

b)

$$\int \frac{1}{x^2 \sqrt{36-x^2}} dx$$

$$x = 6 \cos u$$

$$dx = -6 \sin u \, du$$

$$\sqrt{36-x^2} = \sqrt{36-36\cos^2 u}$$

$$= 6 \sin u$$

$$= \int \frac{1}{36 \cos^2 u \cdot 6 \sin u} (-6 \sin u \, du) \quad \checkmark$$

$$= - \int \frac{1}{36 \cos^2 u} du$$

$$= -\frac{1}{36} \int \sec^2 u \, du \quad \checkmark$$

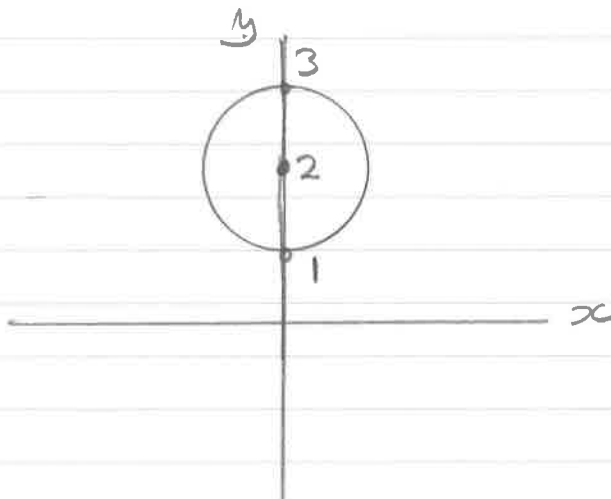
$$= -\frac{1}{36} \tan u + C$$

$$= -\frac{1}{36} \frac{\sqrt{36-x^2}}{x} + C \quad \checkmark$$



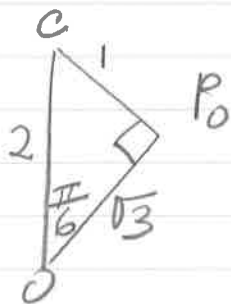
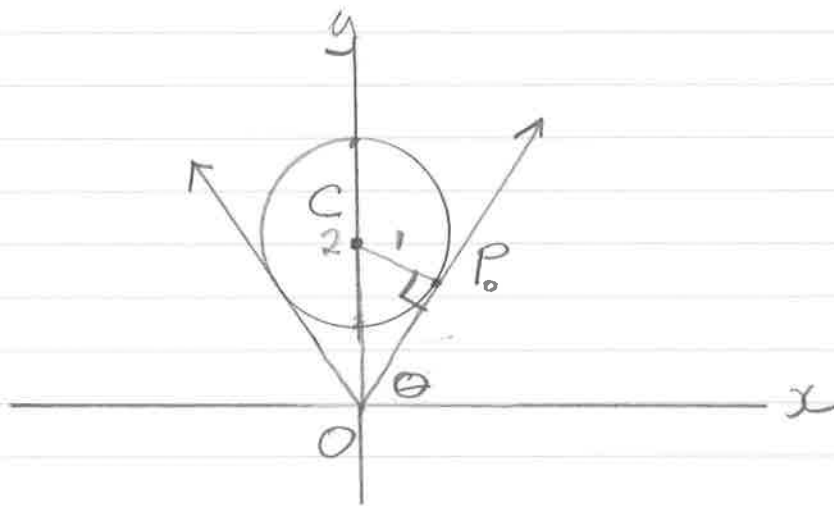
c)

(i) $|z - 2i| = 1$



$\checkmark$ need centre
& radius

ii)



$$\angle COP_0 = \frac{\pi}{6} \quad \checkmark$$

$$\text{so } \theta = \frac{\pi}{2} - \frac{\pi}{6} \\ = \frac{\pi}{3}$$

minimum value of $\arg z$ is $\frac{\pi}{3} \quad \checkmark$

So maximum value of $\arg z$ is $2\pi - \frac{\pi}{3}$

$$= \frac{2\pi}{3} \quad \checkmark$$

iii) At P_0 , $z = \sqrt{3} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \quad \checkmark$

$$= \sqrt{3} \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$$

$$= \frac{\sqrt{3}}{2} + i \frac{3}{2} \quad \checkmark$$

Q11.

$$\begin{aligned} \text{(a)} \quad \int x^2 \cos x \, dx &= x^2 \sin x - \int 2x \sin x \, dx \quad \checkmark \\ &= x^2 \sin x - 2 \left[-x \cos x - \int -\cos x \, dx \right] \\ &= x^2 \sin x + 2x \cos x - 2 \sin x + C \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \text{LHS} &= 3 \times \sin \frac{\theta}{2} \\ &= \sin \frac{\theta}{2} (1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) \\ &= \sin \frac{\theta}{2} + 2 \cos \theta \sin \frac{\theta}{2} + 2 \cos 2\theta \sin \frac{\theta}{2} + \\ &\quad 2 \cos 3\theta \sin \frac{\theta}{2} \\ &= \sin \frac{\theta}{2} + \left(\sin \frac{3\theta}{2} - \sin \frac{\theta}{2} \right) + \left(\sin \frac{5\theta}{2} - \sin \frac{3\theta}{2} \right) \\ &\quad + \left(\sin \frac{7\theta}{2} - \sin \frac{5\theta}{2} \right) \quad \checkmark \\ &= \sin \frac{7\theta}{2} \\ &= \text{RHS} \end{aligned}$$

$$\text{ii)} \quad \text{Let } \theta = 2\pi$$

$$\begin{aligned} \text{Then } \sin \frac{\pi}{7} \left(1 + 2 \cos \frac{2\pi}{7} + 2 \cos \frac{4\pi}{7} + 2 \cos \frac{6\pi}{7} \right) \\ &= \sin \frac{\pi}{7} \left(\frac{2\pi}{7} \right) \\ &= \sin \pi \quad \checkmark \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Now } \sin \frac{\pi}{7} \neq 0 \quad \text{so } 1 + 2 \cos \frac{2\pi}{7} + 2 \cos \frac{4\pi}{7} + 2 \cos \frac{6\pi}{7} \\ &= 0 \end{aligned}$$

iii)

$$\begin{aligned} S &= 1 + 2\cos\theta + 2\cos 2\theta + 2\cos 3\theta \\ &= 1 + 2\cos\theta + 2(2\cos^2\theta - 1) + 2(4\cos^3\theta - 3\cos\theta) \end{aligned}$$

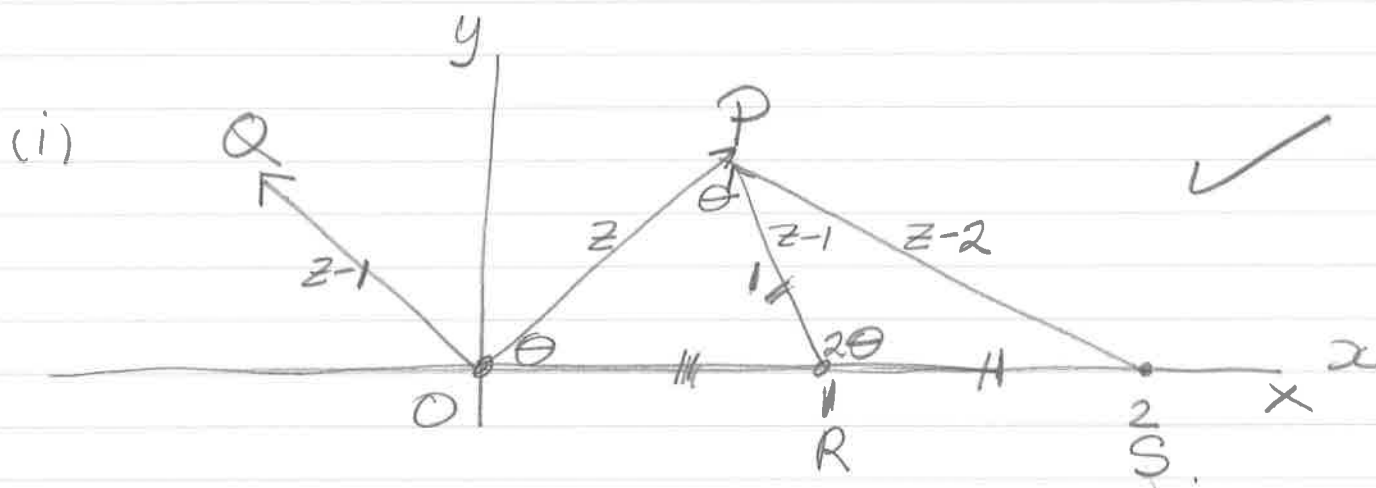
$$\left[\text{aside } \cos 3\theta = \cos\theta \cos 2\theta - \sin\theta \sin 2\theta \right. \\ \left. = \cos\theta(2\cos^2\theta - 1) - 2\cos\theta(1 - \cos^2\theta) \right. \\ \left. = 4\cos^3\theta - 3\cos\theta \right]$$

$$\begin{aligned} \text{So } S &= 8\cos^3\theta + 4\cos^2\theta - 4\cos\theta - 1 \\ &= 0 \quad \text{for } \theta = \frac{2\pi}{7} \end{aligned}$$

$$\begin{aligned} \text{So } x &= \cos \frac{2\pi}{7} \text{ is a solution} \\ \text{of } 8x^3 + 4x^2 - 4x - 1 &= 0 \end{aligned}$$

✓✓

C.



(ii) Call $(1,0)$ R, $(2,0)$ S.

Then $OR = RP = 1$

So $\triangle OPR$ is isosceles

So $\angle ROP = \angle OPR = \theta$

The exterior angle of a triangle equals the sum of the interior opposite angles

So $\angle PRS = 2\theta$

ie $\arg(z-1) = 2\theta$

$OR = RP = RS = 1$, So $\angle OPS$ is an angle in a semi circle

So $\angle OPS = \frac{\pi}{2}$.

So $\angle PSX = \theta + \frac{\pi}{2}$, ext angle of $\triangle OPS$.

ie $\arg(z-2) = \theta + \frac{\pi}{2}$



$$\begin{aligned}
 \text{iii) } \arg\left(\frac{z^2 - 2z}{z-1}\right) &= \arg \frac{z(z-2)}{z-1} \\
 &= \arg z + \arg(z-2) - \arg(z-1) \\
 &= \theta + \theta + \frac{\pi}{2} - 2\theta \\
 &= \frac{\pi}{2}
 \end{aligned}$$

iv) It lies on the upper half of the imaginary axis ✓

v) Let $z = 1 + \cos \alpha$

$$\frac{z(z-2)}{z-1} = \frac{(\cos \alpha + 1)(\cos \alpha - 1)}{\cos \alpha}$$

$$= \frac{\cos 2\alpha - 1}{\cos \alpha} \times \frac{\cos(-\alpha)}{\cos(-\alpha)}$$

$$= \cos \alpha - \cos(-\alpha)$$

$$= 2 \sin \alpha \text{, which has a maximum of 2}$$

So maximum value of $\left|\frac{z^2 - 2z}{z-1}\right|$ is 2.

